

KAYA

"Kaya mo 'to."

An AI budgeting copilot that turns impulse spending into on-chain savings, gasless payments, and real financial credibility — built on Stellar.

APEC x Stellar Hackathon (SEA) · May–July 2026 · Team KAYA · https://github.com/ShinkoSiVes/KAYA_apac-x-stellar_Hackathon

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Practical Payments Meaningful DeFi Localized Tools Composability

1. The Problem (and why it's local)

Young Filipinos are entering the digital economy faster than any generation before them — but the tools around them are built for *tracking*, not *behavior change*, and almost none connect to productive assets.

- **Impulse spending is the default.** E-wallets (GCash, Maya) make spending frictionless and saving an afterthought. A ₱250 coffee, repeated weekly, is the difference between a savings buffer and none.
- **No emergency buffer.** A large share of young Filipinos hold near-zero liquid savings; one emergency means high-interest debt.
- **Thin credit files.** Lenders have almost no data on a 22-year-old's discipline, so responsible young people still get rejected or charged predatory rates.
- **Passive savings.** Even saved money just sits in a wallet earning nothing.

The gap: no tool closes the loop between behavior (coaching at the moment of decision), productive assets (yield on what you don't spend), payments (spending what you saved), and credibility (turning discipline into borrowing power). KAYA closes that loop, and Stellar is the rail that makes it possible.

2. Who it's for

Persona	Situation	What KAYA gives them
Iskolar — student (18–24)	Allowance-based, tempted by food delivery & online sales, zero savings	A coach that says "save it instead," turning saved ₱250 coffees into a growing, yield-earning balance

Yuppie — young pro (23–30)	First real income, lifestyle creep, wants to save but lacks discipline	Automated cooling-off savings, gasless local payments, and a credit reputation for future loans
Negosyante — micro-merchant	Accepts QRPh, wants instant settlement	Receives the asset they want via Stellar path payments when a KAYA user pays

Primary market: Philippines, designed to extend across SEA by swapping in local anchors and stablecoins.

3. The Solution — the KAYA "Money Loop"

Fund In (fiat → stablecoin) → AI Coach ("Kaya afford it?") → Save it (vault) OR Pay it (QR) → On-chain Reputation → Future Credit

1. **Fund in** — deposit local fiat (GCash/Maya/bank) via a Stellar anchor; receive a local stablecoin. **ON/OFF-RAMPS**
2. **AI coaches** every meaningful purchase: "Kaya mo ba 'to?" (Can you afford this?).
3. **Branch: No** → **"Save it Instead"** routes money into a Soroban yield vault with a cooling-off lockup. **Yes** → **"Pay It!"** executes an instant gasless QR path payment.
4. **Reputation** — every disciplined action updates an on-chain Financial Health Score other protocols can read.

4. Feature Deep-Dives

Feature 1 — The "Delayed Gratification" Vault Meaningful DeFi

User story: Maya wants a ₱250 latte. KAYA flags she's over her "wants" budget and offers a "Save it Instead" button. One tap moves ₱250 into her vault — earning micro-yield, locked for 48 hours so the craving passes.

HOW IT WORKS ON STELLAR

- A Soroban smart contract ("KAYA Vault") accepts deposits of a local stablecoin.
- Idle funds are routed into an existing Stellar yield aggregator / liquidity pool (we compose, not build).
- **The twist — a 48-hour cooling-off lockup** enforced per deposit: DeFi mechanics used to break the impulse-spending cycle.

Why it judges well: DeFi with a genuine human purpose tied to a real asset and a real local problem — not yield for yield's sake.

Composability: Soroban contract + third-party yield pool + anchor-issued stablecoin, stitched into one feature.

Feature 2 — Native KAYA Wallet Funding & Cash-Out On/Off-Ramps

User story: Carlo links GCash, taps "Add ₱1,000," confirms in the familiar flow, and seconds later sees ₱1,000 spendable in KAYA. "Cash Out" returns pesos to his bank or e-wallet.

HOW IT WORKS ON STELLAR

- SEP-24 (hosted) and SEP-6 (programmatic) deposit/withdrawal against existing SEA anchors.
- Deposit via GCash, Maya, or bank → anchor issues the local stablecoin straight into the wallet.
- SEP-10 (wallet auth) + SEP-12 (KYC handoff) so KAYA never custodies identity documents.

Why it judges well: the clearest proof of "build on existing rails, not your own gateway." We explicitly consume anchor infrastructure.

Composability: zero payment-gateway code — KAYA is a thin client over standardized anchor SEPs.

Feature 3 — "Kaya Afford It? Pay It!" QR Payments Practical Payments

User story: The AI approves Maya's groceries — "Kaya mo 'to!" A "Pay Now" button appears; she scans the store's QRPh code and the exact amount is paid. She never thinks about crypto, gas, or conversion.

HOW IT WORKS ON STELLAR

- QR scanner compatible with QRPh / regional EMVCo standards.
- A Stellar path payment converts her stablecoin to the merchant's required asset atomically (composing native DEX/AMM liquidity).
- **Fee-bump (sponsored fees):** KAYA's backend pays the microscopic XLM fee → the app feels completely gasless.

Why it judges well: a real purchase, at a real merchant, on a real local QR standard, with UX indistinguishable from GCash.

Composability: native DEX/AMM for conversion + built-in fee-bump for gasless UX. No custom relayer.

Feature 4 — On-Chain Financial Health Score Identity & Credibility

User story: After months of hitting goals and respecting cooling-off periods, Carlo has a verifiable KAYA Score. A Stellar micro-lender later offers a low-collateral loan because it could read his track record on-chain.

HOW IT WORKS ON STELLAR

- Good habits recorded as on-chain attestations / entries in a Soroban reputation registry.
- The score is a portable, user-controlled Web3 primitive.
- Other Stellar micro-lending protocols read it to grant under-collateralized loans.

Why it judges well: extends a simple "health score" into durable, ecosystem-wide utility — exactly what composability unlocks.

Composability: designed to be consumed by others — KAYA becomes a building block in the wider Stellar credit stack.

5. Architecture

Layer	Components
Client	KAYA Mobile App (React Native / Flutter) · AI Budgeting Engine · Stellar Wallet (Freighter / Passkey)
Backend	KAYA Backend orchestrating AI + Stellar SDK · Fee-Bump service that sponsors network fees (gasless UX)
Stellar	Local Anchor (SEP-24/6) · Horizon / Soroban RPC → Soroban Vault (48h lockup + yield), Path Payment (→ QR merchant), Reputation Registry

6. Stellar Tech Stack

Layer	Technology / Standard	Used for
Smart contracts	Soroban	Yield vault w/ cooling-off lockup; reputation registry
Network access	Horizon + Soroban RPC	Submitting tx, reading state
Payments	Path Payments	QR purchase: stablecoin → merchant asset
UX / fees	Fee-Bump (sponsored fees)	Gasless experience for users
Liquidity	Stellar DEX / AMM + yield pool	Conversion + vault yield (composed, not built)

On/off-ramp	SEP-24, SEP-6	Fiat deposit/withdraw via local anchors
Auth / KYC	SEP-10, SEP-12	Wallet auth + anchor-side KYC
Asset	Local PHP / SEA stablecoin	Unit of account throughout
Wallet	Freighter / Passkey smart-wallet	Low-friction key management

7. Composability — reuse vs. build

We build (thin, opinionated): the AI coaching layer, the cooling-off vault logic, the reputation registry schema, and the consumer UX that makes it feel like a normal peso wallet.

We reuse (the heavy lifting): existing SEA anchors (SEP-24/6); existing yield pools + Stellar native DEX/AMM; path payments + fee-bump; Freighter/passkey wallets + on-chain attestations.

8. Criteria Coverage

Requirement	How KAYA meets it
User-facing financial app for real people	Mobile money coach for young Filipinos
Practical payment app	AI-approved QRPh payments, gasless via fee-bump
Meaningful DeFi using real assets	Soroban yield vault on a local stablecoin w/ 48h lockup
Localized financial tool / local economy	GCash/Maya/bank ramps + PHP stablecoin as unit of account
Integrate with local anchors	SEP-24/SEP-6 against SEA Stellar anchors

Give assets utility (earn/swap/disburse)	Vault yield · path-payment swap · QR pay & cash-out
Composability — reuse building blocks	Anchors, yield pools, DEX/AMM, fee-bump, wallets, attestations
Wallets & identity	Freighter/passkey + on-chain reputation score

9. Roadmap

Phase	Window	Milestones
Foundations	May–early Jun 2026	Wallet + testnet account; issue local stablecoin; SEP-24 deposit demo
Core DeFi + Payments	June 2026	Soroban vault w/ 48h lockup + yield; QR path-payment + fee-bump
AI + Reputation	Early–mid Jul 2026	"Kaya afford it?" coaching; reputation registry + score
Polish + Pitch	Late Jul 2026	End-to-end demo; judging submission
Post-hackathon	Aug–Sep 2026	Anchor partnerships; pilot users; micro-lending integration

10. Risks & Honest Gaps

- **PH anchor availability:** production PH fiat anchors are still maturing → the build integrates the testnet/reference anchor (SEP-24/6), wired to swap in live anchors at launch.
- **Yield source maturity:** we integrate the most established available pool; the vault's yield backend is swappable.
- **Regulatory:** KYC delegated to licensed anchors (SEP-12); under-collateralized lending is a future, partner-provided layer that reads our score — KAYA does not underwrite it.

Note for judges: a fully public production PH fiat anchor with open SEP access is still maturing. For the hackathon build we integrate the Stellar Anchor reference/testnet implementation, with UX wired exactly as it will be for live GCash/Maya/bank anchors at launch.

11. Impact, KPIs & The Ask

Impact: KAYA converts everyday impulse spending into savings, yield, and creditworthiness for the most underserved-by-design segment — young Filipinos — using rails that already exist on Stellar.

KPIs: ₱ redirected from impulse spend into the vault per user/month · % of cooling-off periods that end in keeping the savings · active QR payments settled via path payment · KAYA Scores generated and loan offers unlocked.

The ask: technical mentorship from the Stellar ecosystem (Soroban + anchors), introductions to SEA anchor and yield/lending partners for composability, and the chance to compete for the prize pool to fund a pilot.

Appendix — 90-second Demo Script

1. **Fund in:** "Add ₱1,000" → GCash-style flow (testnet anchor) → balance appears in KAYA.
2. **Get coached:** Try to buy a ₱250 coffee → AI: "Hindi pa kaya this week." → tap **Save it Instead** → ₱250 enters the vault, 48h lock + live micro-yield.
3. **Approved purchase:** Scan a grocery QRPh code → AI: "Kaya mo 'to!" → **Pay Now** → path payment settles, no gas shown (fee-bumped).
4. **Reputation:** Open profile → KAYA Score ticked up → "a lending protocol can read this on-chain."